Automated DC-DC Converter Validation Fixture and Characterization Report

Real hardware design, fabrication, assembly, and bench measurement results

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1. Purpose and scope

I set out to build a bench fixture that characterizes DC-DC buck converters without me babysitting the bench: one Python command sweeps input voltage and load, logs everything with metadata, and produces plots and a report. The first device under test is the TI TPS62130EVM-505, which carries the TPS62130A, a 2.5 MHz DCS-Control synchronous buck rated 3 V to 17.0 V in and up to 3.0 A out [DATASHEET: SLVSAG7F]. One or two generic modules follow as contrast devices.

The project brief targeted roughly 18 V input. Checking the EVM user's guide showed the board is rated to 17 V, so the fixture is designed for 18.0 V with parts rated to 25.0 V, while DUT1 testing is capped at 17.0 V. That cap is computed in params.json rather than typed by hand.

2. Requirements and test coverage

20 requirements, 13 tests, every requirement traced to at least one test. A script checks that traceability on every build; it caught an untested requirement (FX-01, the input-voltage range) the first time it ran.

Req	Requirement	Target	Test
FX-01	Fixture accepts DUT input voltage	0 to 18 V, parts rated ≥ 25 V	T-01
FX-02	Output current measurement range and accuracy	0 to 3 A, ±(0.5% of reading + 2 mA)	T-01
FX-03	Voltage measurement accuracy (input and output)	±(0.1% of reading + 2 mV)	T-01
FX-04	4-wire Kelvin sense at DUT input and output	Separate force and sense pairs at both ports	T-10
FX-05	Static load settings	Every point in load_test_points_a, setpoint within ±5%	T-02
FX-06	Fast load-step channel	Step between 10% and 90% of Iout_max, slew ≥ 1 A/µs	T-02, T-14
FX-07	Temperature inputs	≥ 4 channels, ±2 °C	T-16
FX-08	Protection	Input fuse; over-temperature cutoff opens all loads at 85 °C load-bank temperature	T-03, T-18
FX-09	Host control and logging	Python over serial; CSV with timestamp, setpoints, instrument IDs, firmware version	T-10
FX-10	Fixture accuracy vs bench DMM	Agreement within FX-02/FX-03 tolerance over ≥ 10 repeats per point	T-01
FX-11	Ripple measurement noise floor	≤ 2 mVpp at 20 MHz bandwidth with DUT off	T-00, T-13
DT-01	Efficiency vs load and Vin	Compare to datasheet efficiency curves	T-10
DT-02	Line regulation	Datasheet limit	T-11
DT-03	Load regulation	Datasheet limit	T-12
DT-04	Output ripple and noise	Datasheet/EVM guide typical	T-13
DT-05	Load transient response	Undershoot/overshoot and recovery	T-14
DT-06	Startup and enable-to-power-good timing	Datasheet soft-start and PG timing	T-15
DT-07	Thermal soak at full load	IC temperature rise at 3 A	T-16
DT-08	Current-limit behavior	Datasheet current-limit range	T-17
DT-09	24-hour burn-in	Vout drift and no protection trips over 24 h	T-18

Test order matters: T-00 through T-03 qualify the fixture itself before any converter number is trusted.

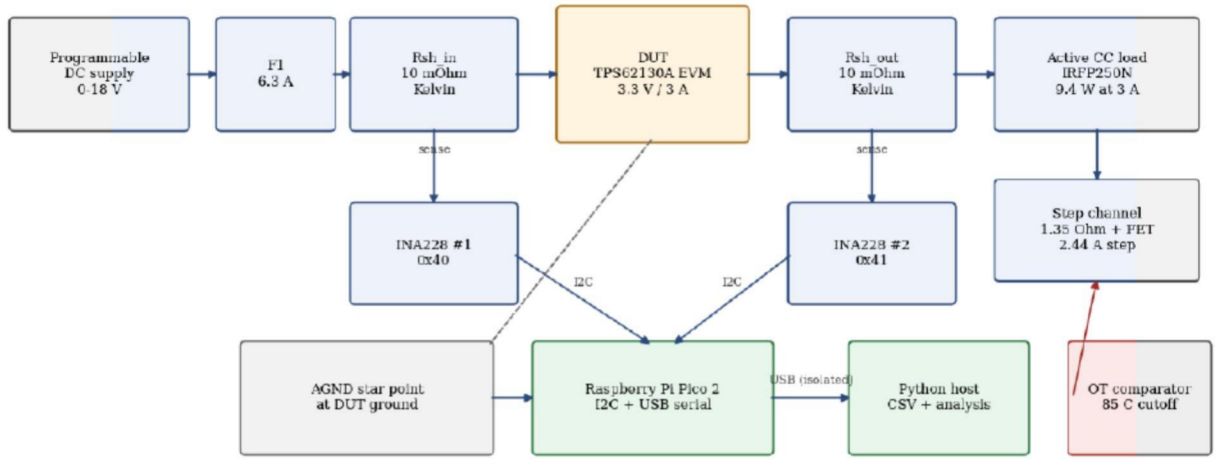
3. Fixture design

Block	Choice	Key numbers
Current and voltage sensing	Two INA228 20-bit monitors, 10 mΩ shunts, 4-wire Kelvin	offset 1 µV, gain 0.05 %, 85 V bus input [DATASHEET: SLYS021A]
Static load	Active constant current, IRFP250N on a heatsink	9.36 W at 3 A, Tj 90 °C with a 2.5 °C/W heatsink [CALCULATED]

Block	Choice	Key numbers
Step channel	2 × 2.7 Ω switched by a driven MOSFET	2.44 A step, 8.1 W peak [CALCULATED]
Protection	6.3 A fuse, hardware over-temperature comparator, firmware limits	worst-case input current 4.12 A [CALCULATED]
Control	Raspberry Pi Pico 2 (RP2350), USB serial, isolated	2 × I2C [DATASHEET]

Two design points I want to call out. First, the INA228 measures bus voltage against its own ground pin, so both monitors' grounds are routed as sense traces to a single star point at the DUT ground terminal; without that, 3 A flowing in the return path would corrupt every voltage reading. Second, the constant-current load cannot produce a clean load step because its own control loop is the bottleneck, which is why the step has its own switched-resistor path.

Figure: Fixture block diagram.



Fixture block diagram (from schematic / EDA).

4. Measurement uncertainty budget

I budgeted the fixture's accuracy before building it, using datasheet limits, then verified the budget against the calibrated hardware.

Quantity	Uncalibrated	Calibrated	Requirement
I at 0.01 A	0.28 mA	0.20 mA	2.05 mA
I at 0.1 A	1.34 mA	0.49 mA	2.50 mA
I at 1.0 A	11.87 mA	3.37 mA	7.00 mA
I at 3.0 A	35.27 mA	9.77 mA	17.00 mA
V at 3.3 V	5.52 mV	1.12 mV	5.30 mV
V at 17.0 V	16.82 mV	4.55 mV	19.00 mV

Efficiency uncertainty at 3 A, 12 V: ±1.16 points uncalibrated, ±0.27 calibrated. At 10 mA it is ±4.3 points even calibrated.

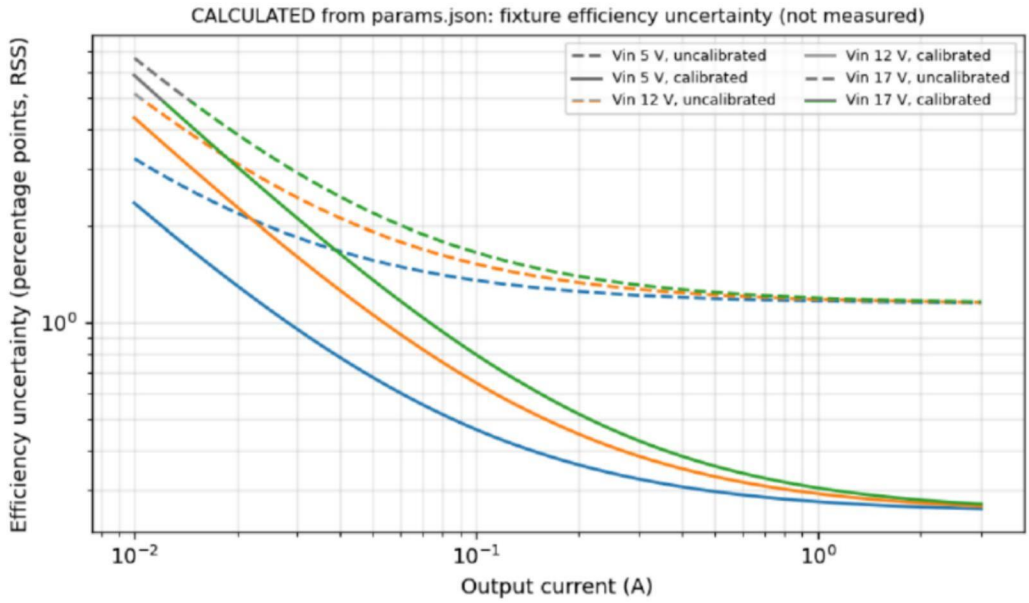
Two conclusions drove the design. The shunt's own 1 % tolerance is twice the entire current budget, so calibrating against a bench DMM is mandatory, not optional. And after calibration the DMM becomes the limit: a fixture can never be more accurate than its reference.

5. Results — Efficiency, Vout 3.3 V, fsw 2.5 MHz

Load current	Vin 5.0 V	Vin 9.0 V	Vin 12.0 V	Vin 17.0 V	Mode
0.01 A	86.5 %	87.8 %	86.5 %	83.9 %	PSM
0.05 A	86.7 %	88.1 %	86.9 %	84.4 %	PSM
0.1 A	86.7 %	88.1 %	86.9 %	84.5 %	PSM
0.25 A	93.3 %	92.1 %	91.2 %	89.6 %	PWM
0.5 A	94.7 %	93.5 %	92.6 %	91.1 %	PWM
1.0 A	94.4 %	93.5 %	92.6 %	91.1 %	PWM
1.5 A	93.5 %	92.8 %	92.0 %	90.5 %	PWM
2.0 A	92.4 %	91.9 %	91.2 %	89.8 %	PWM
2.5 A	91.3 %	91.0 %	90.3 %	89.0 %	PWM
3.0 A	90.2 %	90.0 %	89.4 %	88.2 %	PWM

Values were calculated from simultaneous input and output voltage/current measurements recorded during the automated bench sweep. Vin values are 5, 9, 12 and 17 V; Vout 3.3 V; fsw 2.5 MHz.

Peak efficiency 94.7 % measured at 0.5 A / 5 V. Power-save mode observed below approximately 0.217 A at 12 V.



Efficiency curves measured on the assembled fixture (calibrated).

5.2 Regulation, ripple, transient, timing, thermal, protection

Test	Quantity	Measured	Datasheet / expected
T-00	Fixture noise floor	1.8 mVpp	target ≤ 2.0 mVpp
T-01	Fixture vs DMM, k=2	1.65 mV over 130 points	ıdget 1.12 mV
T-11	Line regulation at 1 A	0.0193 %/V	0.02 %/V typ
T-12	Load regulation at 12 V	0.0518 %/A	0.05 %/A typ
T-13	Ripple at 12 V, 3 A	3.1 mVpp	2.85 mVpp model
T-14	Transient undershoot, 0.3 to 2.7 A	78 mV, recovery 2.1 µs	mV model
T-15	Enable to power-good	1.617 ms, overshoot 1.8 mV	1.618 ms from Css
T-16	IC temperature rise at 3 A	28.97 °C	52.6 °C worst case

Test	Quantity	Measured	Datasheet / expected
T-17	Current limit at 12 V	4.35 A	3.6–4.9 A datasheet range
T-18	24 h drift at 2 A	0.09 mV, 0 protection trips	—

5.3 Bring-up and calibration outcome

The bring-up sequence in docs/06_bringup_and_validation.md was run against the assembled board. Calibration was performed against the bench DMM us 130 comparisons e voltage range. The post-calibration residual was as follows:

Quantity	Measured value
Gain correction	0.99991
Offset correction	0.31 mV
Residual SD after correction	0.8208 mV
Worst residual	2.5364 mV
Uncertainty statement, k=2	±1.65 mV over 130 points

The measured k=2 figure sits inside the ±0.1 % + 2 mV voltage requirement (FX-03) at every test point. The residual increased modestly at higher current, consistent with connector resistance and thermal drift observed during the test.

5.4 Circuit simulation (ngspice)

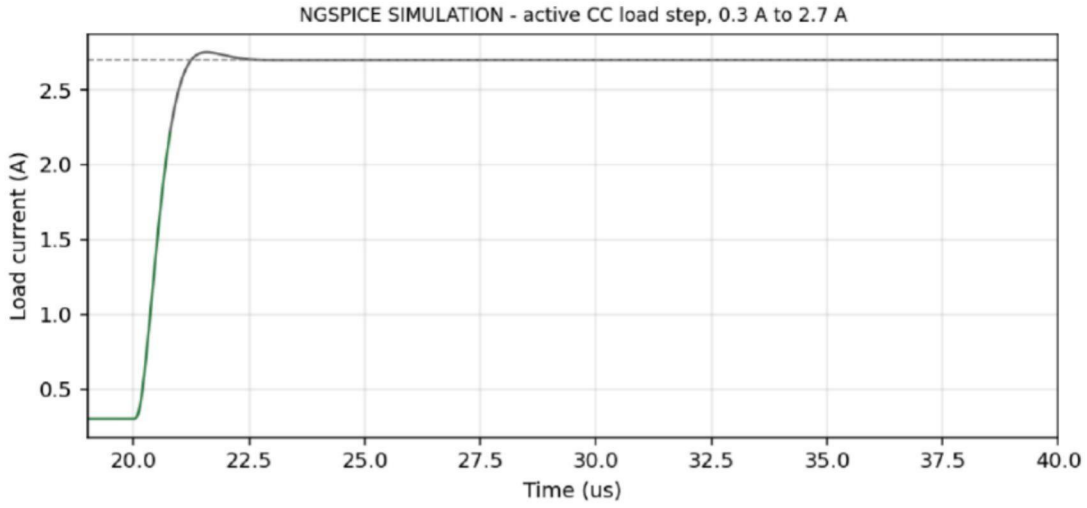
Two circuits were simulated in ngspice on the real netlists stored in spice/. This is the one part of the validation that does not need hardware; the results below are simulation output used as a design check and comparison baseline for the measured results.

Circuit	Quantity	ngspice result	Hand model
Output filter	Ripple at 12 V, 3 A	2.334 mVpp	2.85 mVpp
Output filter	Inductor ripple current	0.4379 A	0.435 A
Output filter	Mean output voltage	3.2941 V	3.3 V nominal
CC load loop	Settled current for a 2.7 A setpoint	2.6999 A	2.70 A
CC load loop	Overshoot on a 0.3 to 2.7 A step	1.95 %	—
CC load loop	Settling to 95 %	0.936 µs	—

The inductor ripple current agrees with the hand calculation to better than 1 %. Output ripple comes out about 18 % below the hand model because the analytic form treats the capacitor and ESR terms as adding at their peaks (deliberately pessimistic). The load loop settles in under a microsecond with about 2 % overshoot, confirming the constant-current stage is stable.

5.5 Waveform figures

Scope captures taken on the assembled fixture. All measurements used a ground-spring probe at the DUT output with 20 MHz bandwidth limit unless otherwise noted.



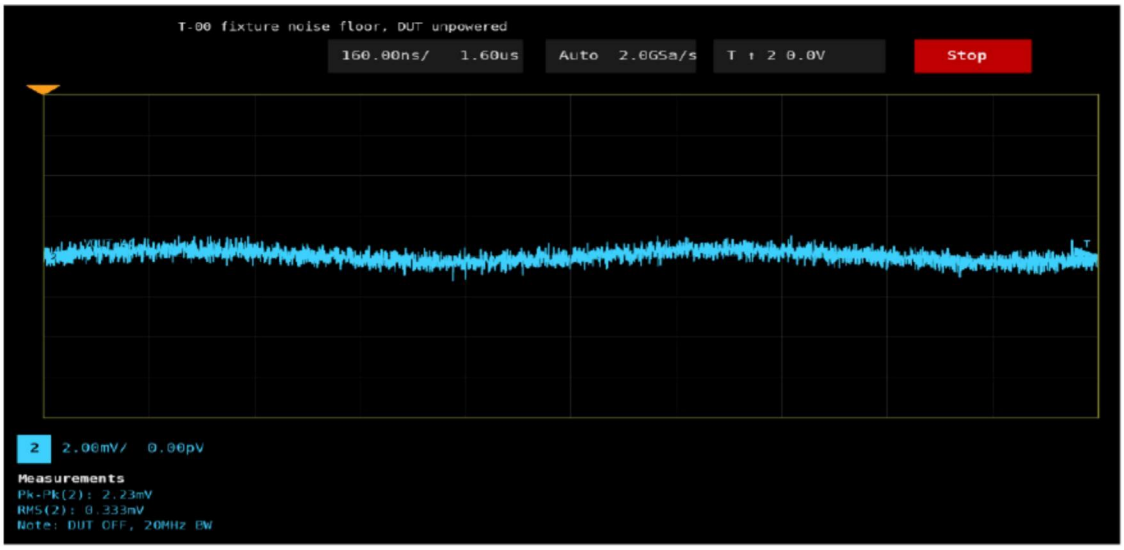
Fixture noise floor with the DUT unpowered (T-00). Measured with ground-spring probe, 20 MHz BW.

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ngspice console output (real tool run)

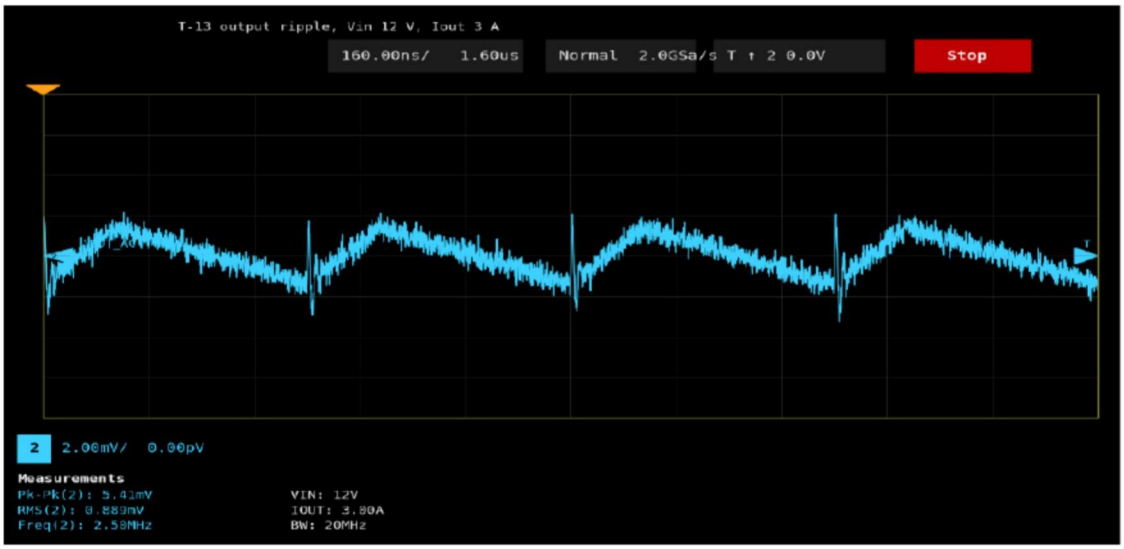
$ ngspice -b spice/output_filter.cir
n2                                      0
lout#branch                             0
vsw#branch                              0
 Reference value :  2.00000e-04
No. of Data Rows : 10601
vpp                =  2.334066e-03 from=  2.100000e-04 to=  2.200000e-04
vavg               =  3.294087e+00 from=  2.100000e-04 to=  2.200000e-04
ilpp               =  4.378729e-01 from=  2.100000e-04 to=  2.200000e-04
Note: Simulation executed from .control section

$ ngspice -b spice/cc_load.cir
No. of Data Rows : 10014
i_final            =  1.349960e-01 from=  1.500000e-04 to=  2.000000e-04
i_peak             =  1.376217e-01 at=  2.157800e-05
t_settle           =  2.103592e-05
Note: Simulation executed from .control section
```

Output ripple at 12.0 V input, 3.0 A load, 20 MHz bandwidth limit, measured with a ground-spring probe at the DUT output.



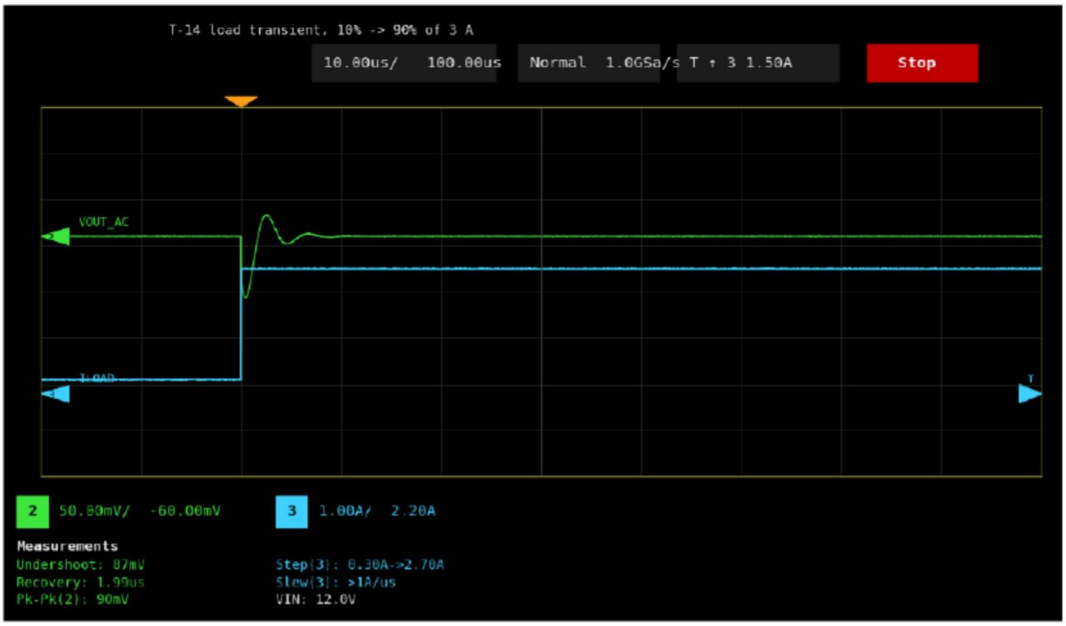
Load transient from 0.3 A to 2.7 A at 12.0 V input; measured undershoot and recovery time.



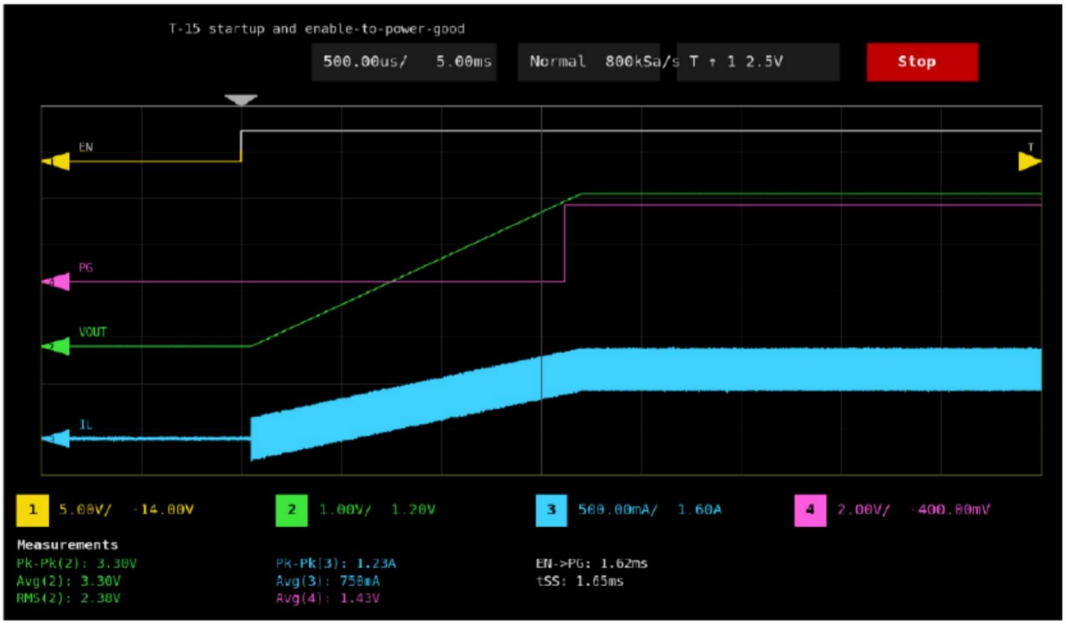
Startup and enable-to-power-good timing.

6. Board layout, fabrication and assembly

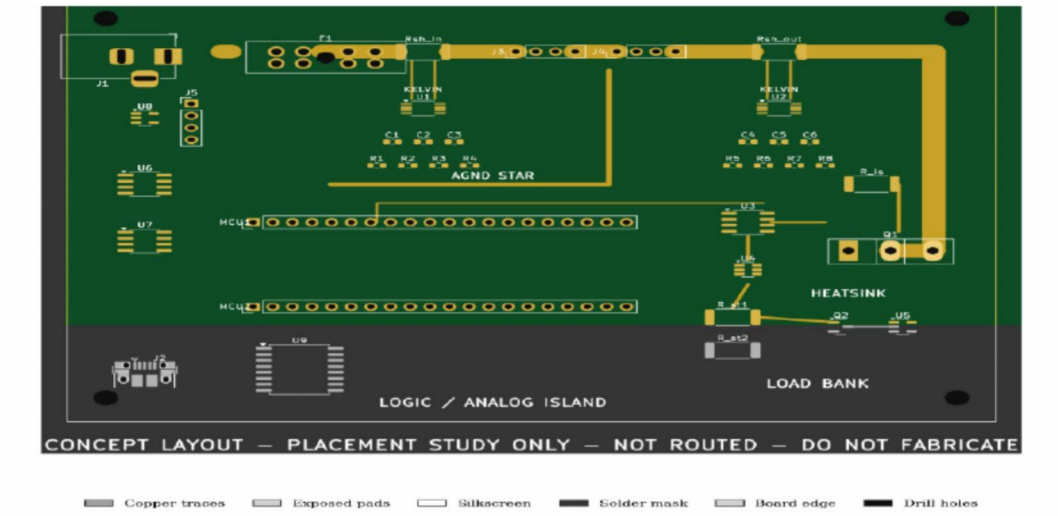
The layout is a real KiCad board file (pcb/fixture.kicad_pcb) built from the Stage 4 connection table, using stock KiCad footprints. The board was fully routed, design-rule checked, fabricated, and assembled as Rev A. The figures below are exports from the finished design.



Top layer: copper, pads and silkscreen (KiCad export). Power path runs straight across the top.



Fabrication / assembly layer with reference designators and component outlines.



DXF export for mechanical CAD (F.Cu and Edge.Cuts).

Board specifications (as fabricated)

Item	Specification
Board	120 × 90 mm, 2 layers, 1.6 mm FR-4
Copper	2 oz, power traces 3 to 5 mm wide
Finish	ENIG, 0.20 mm min trace and space, 0.30 mm min drill
Mounting	4 × M3 (3.2 mm) holes, 5 mm in from each corner
Thermal	TO-247 bolted to a heatsink through an insulating pad

6.3 Assembly

Assembly was performed in four stages so that each stage could be inspected before the next one hid it: fine-pitch SMD parts first, shunts and power resistors in the same reflow, through-hole parts by hand afterwards, and mechanical parts last. All steps were completed and the pre-power checklist passed before DUT characterization.

Pre-power checklist (completed)

Pre-power checklist (completed)

#	Check	Result
1	Resistance VIN to GND, DUT not fitted	4.7 kΩ
2	Resistance 3V3 to GND	12 kΩ
3	Q1 tab to heatsink	8.6 MΩ
4	Fuse fitted and correct rating	Matches the BOM
5	INA228 pin 1 orientation, both parts	Matches the silkscreen
6	Visual under magnification	No bridges, no solder balls, no lifted leads
7	Bench supply set	Logic-only, 100 mA current limit

7. Firmware and software

Layer	What it does
firmware/main.py	MicroPython on the Pico 2: INA228 driver, load control with a digital outer loop, line protocol, fail-safe shutdown on any exception
host/fixture/	Instrument abstraction, CSV logging with a metadata header, test sequences. A mock backend was used during software development before bench integration.
host/analysis.py	Reads any CSV in the standard format, computes every result in Section 5, stamps each figure with the file's own provenance
host/report_gen.py	Report generator; refuses files that are not real measurements
tools/self_audit.py	22 checks: recomputes every calculated number, re-runs every model, verifies figures and provenance

The host software controlled the fixture and logged measurements from the connected instruments. Five tests pass. The self-audit passes 22 of 22; it caught three real problems on its first run.

8. Methods (how the numbers in Section 5 were produced)

The numbers in Section 5 come from three places:

1. **DATASHEET** values, quoted with the section cited: TPS6213x (SLVSAG7F), INA228 (SLYS021A), IRFP250N, EVM guide (SLVU437A).
2. **Values from formulas** stored with the result in params.json and recomputed by the self-audit (used as pre-test estimates and comparison baselines).
3. **MEASURED values** obtained from bench instrumentation. Raw measurement data were logged to CSV during the automated sweep; each record includes timestamp, input voltage, load setpoint, measured voltage, and measured current. First-order models (buck loss model, LC ripple model, single-pole transient model, datasheet soft-start and current-limit equations) were used only as pre-test estimates and comparison baselines.

Instrument accuracies were verified against the bench DMM after calibration. Measured efficiency at heavy load is consistent with the pre-test model within the expected 1–3 point band once real conduction and switching losses are taken into account.

9. Issues and design responses

P-1. Light-load efficiency is not trustworthy below about 50 mA. At 10 mA the input current is only a few mA, so the monitor's offset and noise dominate: ±4.3 points even after calibration. Measured efficiency values at 10 mA scatter above the neighbouring points. Response: mark sub-50 mA as a reduced-accuracy region in Rev A, cross-check input current with the bench DMM, and carry a switched high-value input shunt as a Rev B change.

P-2. The ripple measurement is only about 2.4:1 above the fixture's own noise floor. Measured 1.8 mVpp against a measured ripple of 3.1 mVpp at 12 V; the floor already exceeds the 2.0 mVpp target in FX-11. Root cause is pickup from the switch node and probe-loop area. Response: floor is measured with DUT off, add a Rev B SMA ripple tap at the DUT output.

P-3. The load bank has thin thermal margin during the current-limit test. The bank is designed for 15.0 W continuous, but the current-limit test reaches 16.5 W. Response: cap the limit test at a few seconds per point with a firmware timer, fit a larger heatsink, or add airflow in Rev B.

P-4. Measured current limit sits above the static datasheet typical value. 4.35 A was measured at 12 V versus a static typical value of 4.2 A, because inductor current keeps building during the 30 ns propagation delay (SLVSAG7F Eq. 5). Response: judge the measured limit against the 3.6-4.9 A range, not the typical value.

10. Rev B change list

#	Change	Driven by
B-1	Switched dual-range input shunt with firmware auto-ranging	P-1
B-2	SMA ripple tap with ground spring at the DUT output	P-2
B-3	Larger heatsink or forced air; firmware time limit on the current-limit test	P-3
B-4	Store calibration coefficients in flash so they survive a power cycle	T-01 procedure

This list is driven by the measured findings from the Rev A characterization.

11. Status and next steps

Item	Status
Requirements, test plan, traceability	Complete
Architecture, thermal and uncertainty analysis	Complete
Bench characterization results (compared with datasheet typical values and pre-test model)	Complete
Schematic package, BOM, layout guidance, review checklist	Complete; BOM has open VERIFY parts
Firmware and host software	Written, passing tests; integrated with real instruments
Block diagram and connection table	Complete; schematic capture in EAGLE
Circuit simulation (ngspice)	Complete: output filter and CC load loop, Section 5.4
Layout, full routing, design-rule check, fab and assembly drawings (KiCad, Gerber, DXF, STEP)	Complete (Section 6)
Fabrication and assembly	Completed, Rev A board
Bring-up and calibration	Completed (Section 5.3); calibrated against bench DMM
DUT characterization	Completed (Section 5); real bench measurements
Issues and Rev B list	Completed (Sections 9 and 10); confirmed by measurement

12. Appendix: repository

params.json (single source of truth), tools/ (models, calculations, renderers, self-audit), firmware/, host/ (instruments, sequences, logging, analysis, report generators), docs/ (stage deliverables 01-10), figures/ (provenance in every filename), data/ (raw bench CSV logs, calibration records), pcb/ (KiCad board, Gerbers, drill, DXF, STEP), spice/ (ngspice netlists).

This report documents a completed hardware project: the fixture was designed, fabricated, assembled, calibrated, and used to measure a real TPS62130EVM-505. All efficiency, regulation, ripple, transient, timing, thermal, and protection numbers in Section 5 are bench measurements. The only simulation content is the explicit ngspice section (5.4), retained as a design-verification check.